

1. Smallest Element in Array

Question:

Given an array of integers, find the smallest element present in the array.

Input:

- First line contains an integer N representing the size of array.
- Second line contains N space-separated integers.

Output:

Print the smallest element.

Constraints:

$$1 \leq N \leq 10^5$$

$$-10^9 \leq \text{arr}[i] \leq 10^9$$

Sample Input:

5
12 3 45 1 9

Sample Output:

1

2. Largest Element in Array

Question:

Given an array of integers, find the largest element.

Input:

First line contains N .

Second line contains N integers.

Output:

Print the largest element.

Constraints:

$$1 \leq N \leq 10^5$$

Sample Input:

5
12 3 45 1 9

Sample Output:

45

3. Second Smallest Element**Question:**

Find the second smallest distinct element in the array.

Input:

First line contains N.

Second line contains N integers.

Output:

Print second smallest element.

Constraints:

$$2 \leq N \leq 10^5$$

Sample Input:

6
4 2 8 1 9 3

Sample Output:

2

4. Second Largest Element

Question:

Find the second largest distinct element in the array.

Input:

First line contains N.

Second line contains N integers.

Output:

Print second largest element.

Constraints:

$$2 \leq N \leq 10^5$$

Sample Input:

6
4 2 8 1 9 3

Sample Output:

8

5. Reverse Array**Question:**

Reverse the given array.

Input:

First line contains N.

Second line contains N integers.

Output:

Print reversed array.

Constraints:

$$1 \leq N \leq 10^5$$

Sample Input:

5
1 2 3 4 5

Sample Output:

5 4 3 2 1

6. Frequency of Elements

Question:

Count the frequency of each element in the array.

Input:

First line contains N.

Second line contains N integers.

Output:

Print each element with its frequency.

Constraints:

$1 \leq N \leq 10^5$

Sample Input:

6
1 2 2 3 1 2

Sample Output:

1 2
2 3
3 1

7. Remove Duplicates

Question:

Remove duplicate elements from the array while maintaining insertion order.

Input:

First line contains N.

Second line contains N integers.

Output:

Print array after removing duplicates.

Constraints:

$$1 \leq N \leq 10^5$$

Sample Input:

7

1 2 2 3 4 4 5

Sample Output:

1 2 3 4 5

8. Rotate Array by K Positions**Question:**

Rotate the array to the right by K positions.

Input:

First line contains N.

Second line contains N integers.

Third line contains K.

Output:

Print rotated array.

Constraints:

$$1 \leq N \leq 10^5$$

Sample Input:

5

1 2 3 4 5

2

Sample Output:

4 5 1 2 3

9. Find Repeating Elements**Question:**

Find all repeating elements in the array.

Input:

First line contains N.

Second line contains N integers.

Output:

Print repeating elements.

Constraints:

$1 \leq N \leq 10^5$

Sample Input:

7

1 2 3 2 4 1 5

Sample Output:

1 2

10. Find Non-Repeating Elements**Question:**

Find all non-repeating elements in the array.

Input:

First line contains N.

Second line contains N integers.

Output:

Print non-repeating elements.

Constraints:

$$1 \leq N \leq 10^5$$

Sample Input:

7
1 2 3 2 4 1 5

Sample Output:

3 4 5

11. Equilibrium Index**Question:**

Find the equilibrium index where left sum equals right sum.

Input:

First line contains N.

Second line contains N integers.

Output:

Print equilibrium index, else print -1.

Constraints:

$$1 \leq N \leq 10^5$$

Sample Input:

5
1 3 5 2 2

Sample Output:

2

12. Maximum Product Subarray

Question:

Find maximum product of contiguous subarray.

Input:

First line contains N.

Second line contains N integers.

Output:

Print maximum product.

Constraints:

$$1 \leq N \leq 10^5$$

Sample Input:

5
2 3 -2 4 -1

Sample Output:

48

13. Median of Array**Question:**

Find the median of array.

Input:

First line contains N.

Second line contains N integers.

Output:

Print median.

Constraints:

$$1 \leq N \leq 10^5$$

Sample Input:

5
1 3 2 4 5

Sample Output:

3

14. Array Subset Check

Question:

Check whether second array is subset of first array.

Input:

First line contains N1.
Second line contains N1 integers.
Third line contains N2.
Fourth line contains N2 integers.

Output:

Print Yes or No.

Constraints:

$1 \leq N \leq 10^5$

Sample Input:

5
1 2 3 4 5
3
2 4 5

Sample Output:

Yes

15. Search Element

Question:

Search whether a given element exists in array.

Input:

First line contains N.

Second line contains N integers.

Third line contains target element.

Output:

Print Found or Not Found.

Constraints:

$$1 \leq N \leq 10^5$$

Sample Input:

```
5
10 20 30 40 50
30
```

Sample Output:

Found

16. Symmetric Pairs**Question:**

Find symmetric pairs in given input.

Input:

First line contains number of pairs N.

Next N lines contain pair values.

Output:

Print symmetric pairs.

Constraints:

$$1 \leq N \leq 10^5$$

Sample Input:

```
4
1 2
```

3 4
2 1
5 6

Sample Output:

1 2

17. Missing Number

Question:

One number is missing from 1 to N. Find the missing number.

Input:

First line contains N.

Second line contains N-1 integers.

Output:

Print missing number.

Constraints:

$1 \leq N \leq 10^5$

Sample Input:

5
1 2 4 5

Sample Output:

3

18. Leaders in Array

Question:

Find all leader elements in the array.

Input:

First line contains N.

Second line contains N integers.

Output:

Print leader elements.

Constraints:

$$1 \leq N \leq 10^5$$

Sample Input:

6
16 17 4 3 5 2

Sample Output:

17 5 2

19. Pair Sum**Question:**

Find pair whose sum equals target.

Input:

First line contains N.

Second line contains N integers.

Third line contains target sum.

Output:

Print valid pair.

Constraints:

$$1 \leq N \leq 10^5$$

Sample Input:

5
1 4 5 6 2
7

Sample Output:

1 6

20. Union and Intersection**Question:**

Find union and intersection of two arrays.

Input:

First line contains N1.

Second line contains first array.

Third line contains N2.

Fourth line contains second array.

Output:

Print union and intersection.

Constraints:

$1 \leq N \leq 10^5$

Sample Input:

3
1 2 3
3
2 3 4

Sample Output:

Union: 1 2 3 4
Intersection: 2 3

21. Maximum Sum Subarray**Question:**

Find the maximum sum contiguous subarray.

Input:

First line contains N.

Second line contains N integers.

Output:

Print maximum sum.

Constraints:

$$1 \leq N \leq 10^5$$

Sample Input:

8

-2 -3 4 -1 -2 1 5 -3

Sample Output:

7